$$\xrightarrow[\times 1/5]{Row3 = Row3} \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-3\times Row3]{Row1 = Row1} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-4\times Row3]{Row2 = Row2}$$

$$\begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times 1/2]{Row2 = Row2} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-3\times Row2]{Row1 = Row1} \begin{bmatrix} 1 & 0 & 3/2 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in reduced row echelon form

1. Row 1, 2 and 3 are non-zero rows. Row 4 is a zero row. ∴ The condition that all non-zero rows must be above the zero rows is True.
2. Row 1 Leading element = 1 in Column 1
   Row 2 Leading element = 1 in Column 2
   Row 3 Leading element = 1 in Column 4
   ∴ The condition that each row has its leading element in a column to the right of the columns of the leading elements of all rows above it is True.
3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
   Row 3 Column 2 = Row 4 Column 2 = 0
   Row 4 Column 4 = 0
   ∴ The condition that for each column having a leading entry, all positions below the leading entry must contain only 0 is True.
4. The condition that each leading entry is 1 is True.
5. Row 1 column 4 = Row 2 column 4 = 0
   Row 1 Column 2 = 0
   ∴ The condition that for each column having a leading entry, all positions above the leading entry must contain only 0 is True.

So, the matrix $\begin{bmatrix} 1 & 0 & 3/2 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in reduced row echelon form. ◻

∴ It's easy to see that the 1st 3 rows only contain a pivot position.

Since, there does not exist a pivot position in every row, the equation $Ax = b$ does not have a solution for each b in R^4 from Theorem 4 ◻